

The Upper Limb

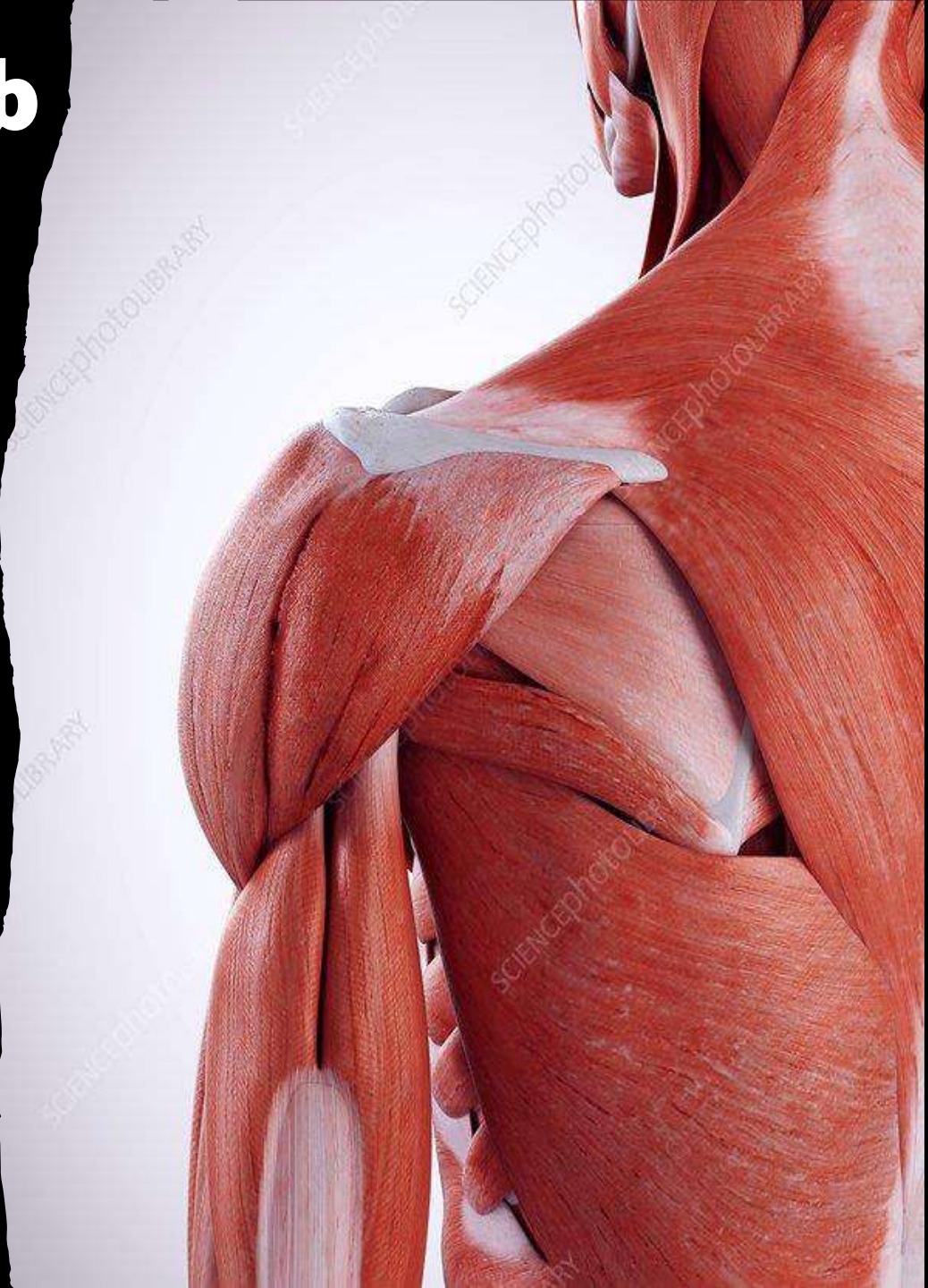
Back

Scapular

Shoulder

Regions

Dr. Rasha Alshali



The shoulder region

SHOULDERS

Deltoid muscle

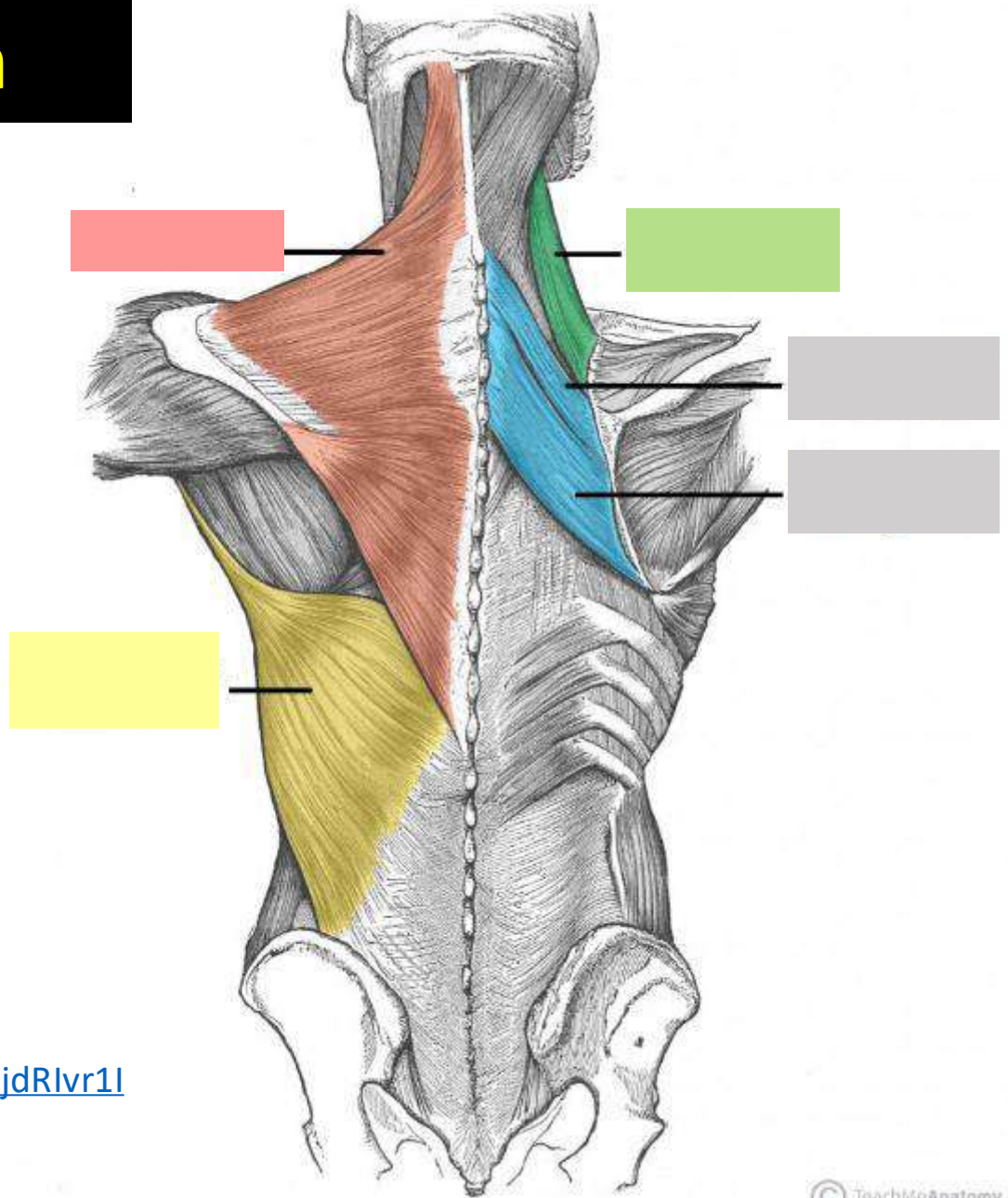
ANTERIOR HEAD

LATERAL HEAD

POSTERIOR HEAD



The Back region

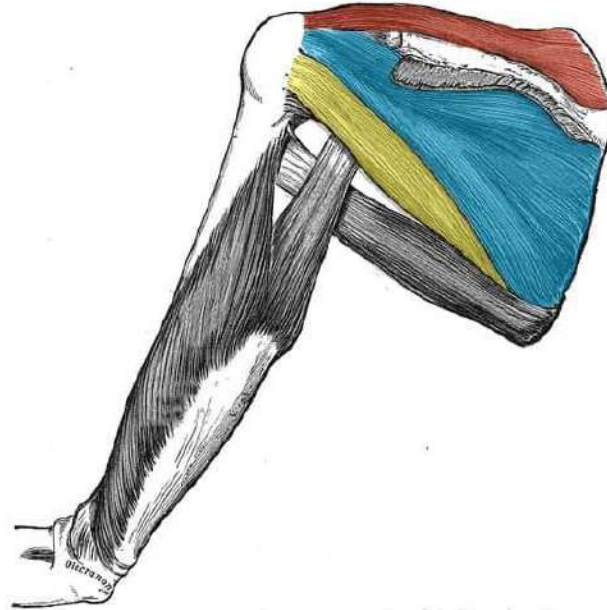
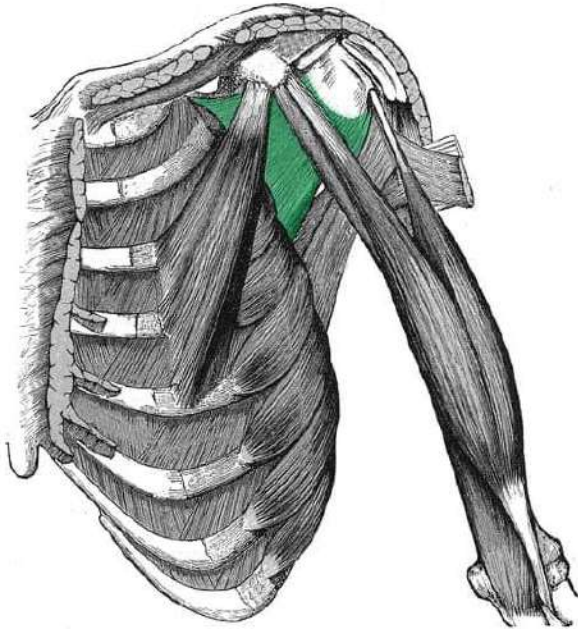


<https://www.youtube.com/watch?v=SOKjdRIvr1I>

Min 0-4

The scapular region

The muscles connecting the scapula to the humerus:



<https://www.youtube.com/watch?v=Ralt79pPfgE>

Min 1-6

<https://www.youtube.com/watch?v=BHemdCT9KLY>

Min 0-2:50

Contents of the Lecture:

The skin of the Back.

Muscles of the **Back** region.

Muscle of the **Shoulder** region.

Muscles of the **Scapular** region.

Rotator cuff muscles

Nerves:

- The axillary nerve
- The suprascapular nerve.

Skin of the Back regions

➤ Arterial supply:

branches from the posterior intercostal arteries & lumbar arteries.

➤ Venous Drainage:

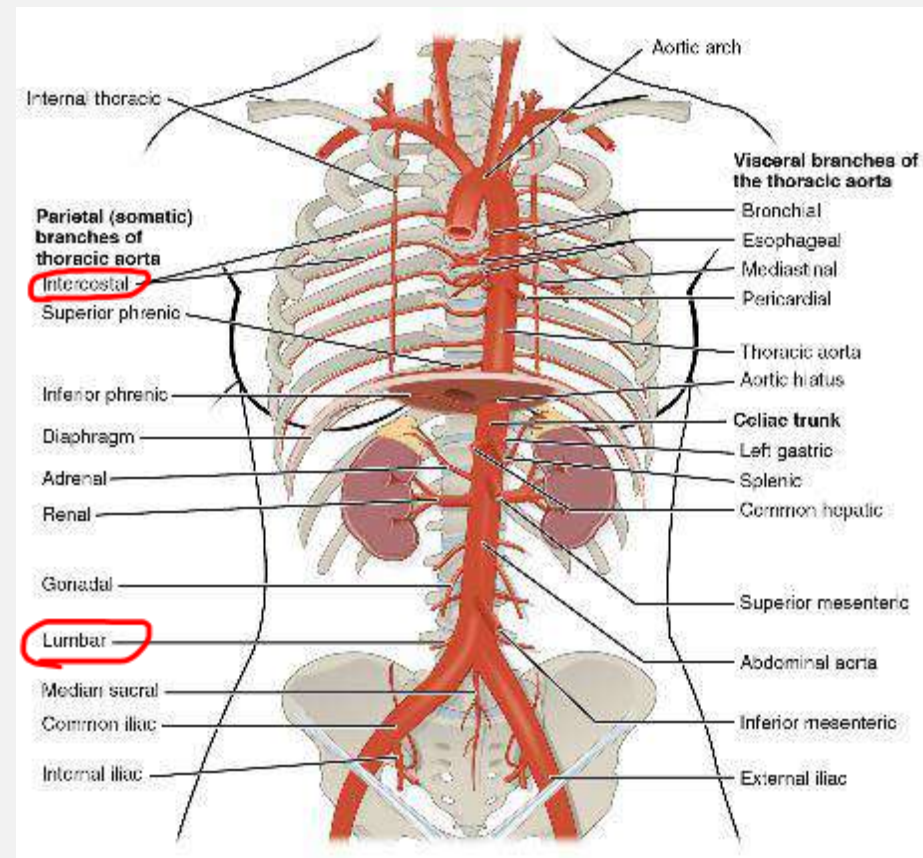
to azygos vein and inferior vena cava.

➤ Nerve supply:

by the posterior rami of the spinal nerves.

➤ Lymph Drainage:

to 'posterior group of axillary LN.
(Subscapular LN)



Muscles of the Back regions

Three Muscle Layers:

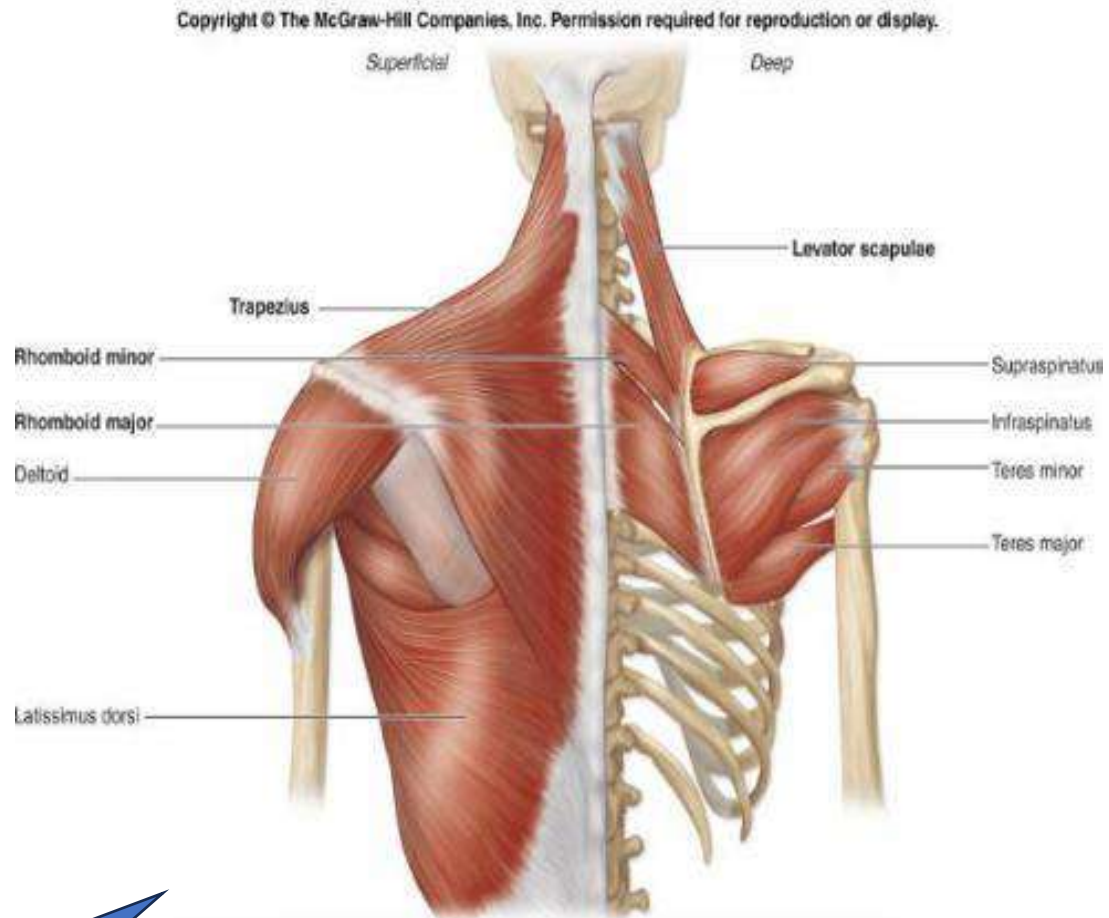
Superficial layer:

- ✓ Trapezius
- ✓ Latissimus dorsi.
- ✓ Levator scapulae
- ✓ Rhomboid minor
- ✓ Rhomboid major

Intermediate layer

Deep layer

Why do we concern about those back muscle?



We are concerned about the **superficial(1st layer)** because these muscles move the upper limb.

Trapezius

It is the upper large triangular muscle

Origin:

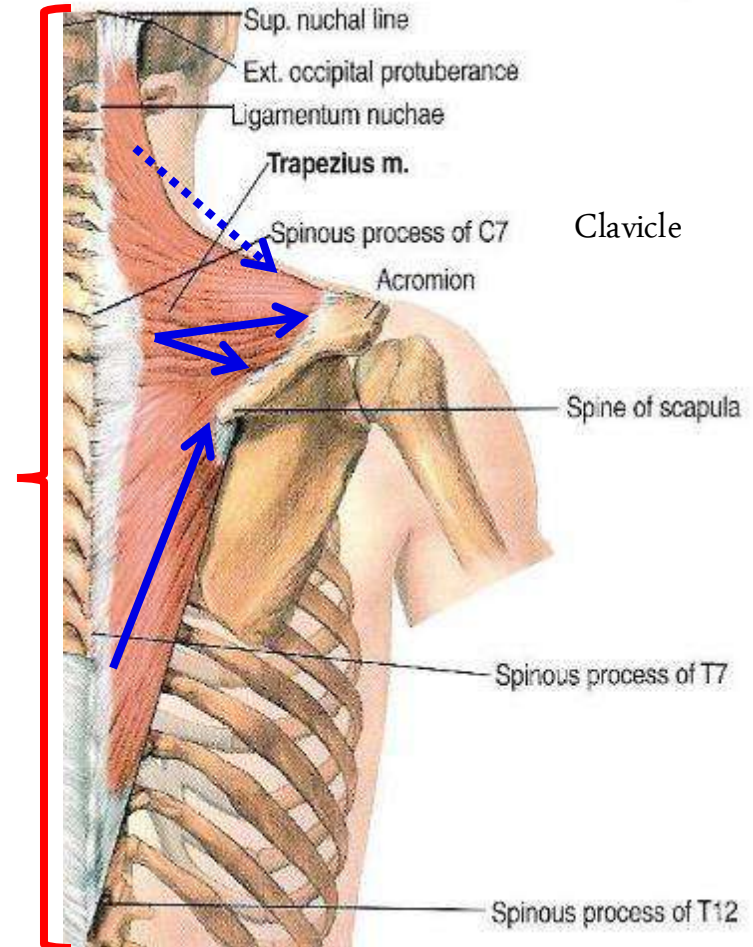
1. Medial third of superior nuchal line
2. External occipital protuberance.
3. Ligamentum nuchae.
4. Spine of C7
5. Spines and supraspinous ligaments of all thoracic vertebrae T1-T12

Insertion:

Upper fibers into → the posterior aspect of lateral third of the **clavicle**.

Middle fibers → the medial aspect of the **acromion** process and upper border of **spine** of the **scapula**.

Lower fibers → medial end of the **spine** of the **scapula**.



Trapezius

Nerve Supply:

Motor fibers from the **spinal part of accessory nerve (11th CN)**.
(cranial nerve)

Sensory fibers from the **C3 & C4 cervical nerves**.
(spinal nerves)

including pain & proprioception

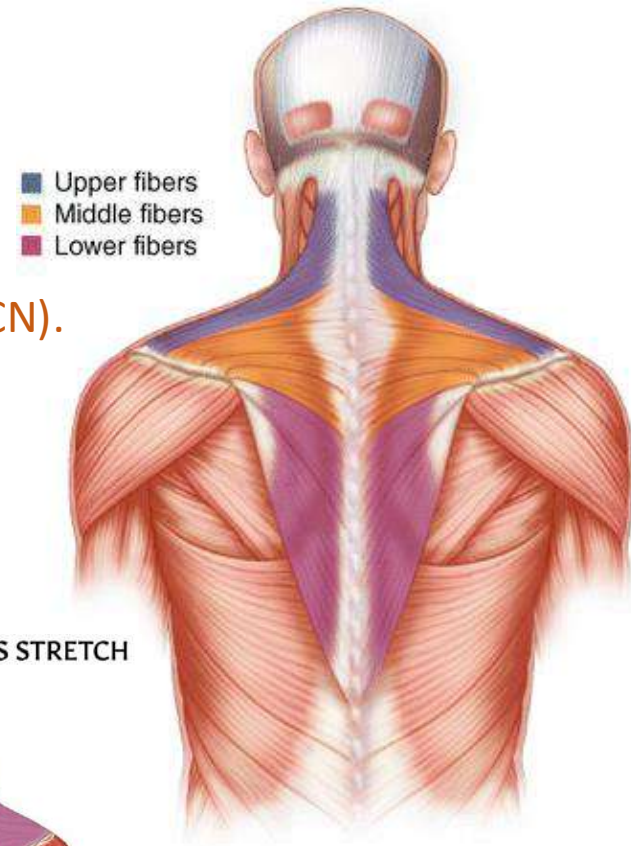
Action:

Upper fibers: elevates the scapula.

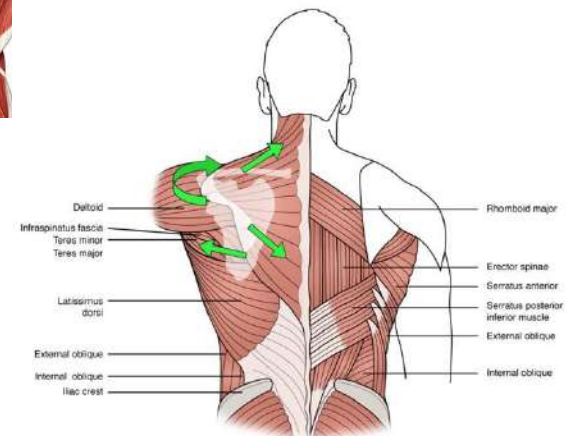
Middle fibers: retracts the scapula

Lower fibers: depresses the scapula.

With serratus anterior: they rotate the scapula upward
(for abduction of the arm > than 90°) above the head



UPPER TRAPEZIUS STRETCH



Latissimus Dorsi

It is the lower large triangular muscle

Origin:

1. posterior part of iliac crest.
2. Lumbar fascia
3. Spines of the **lower** 6 thoracic vertebrae.
4. **Lower** 3 or 4 ribs.
5. **lower** angle of the scapula.

Insertion:

Floor of the bicipital groove of the humerus.

Nerve Supply:

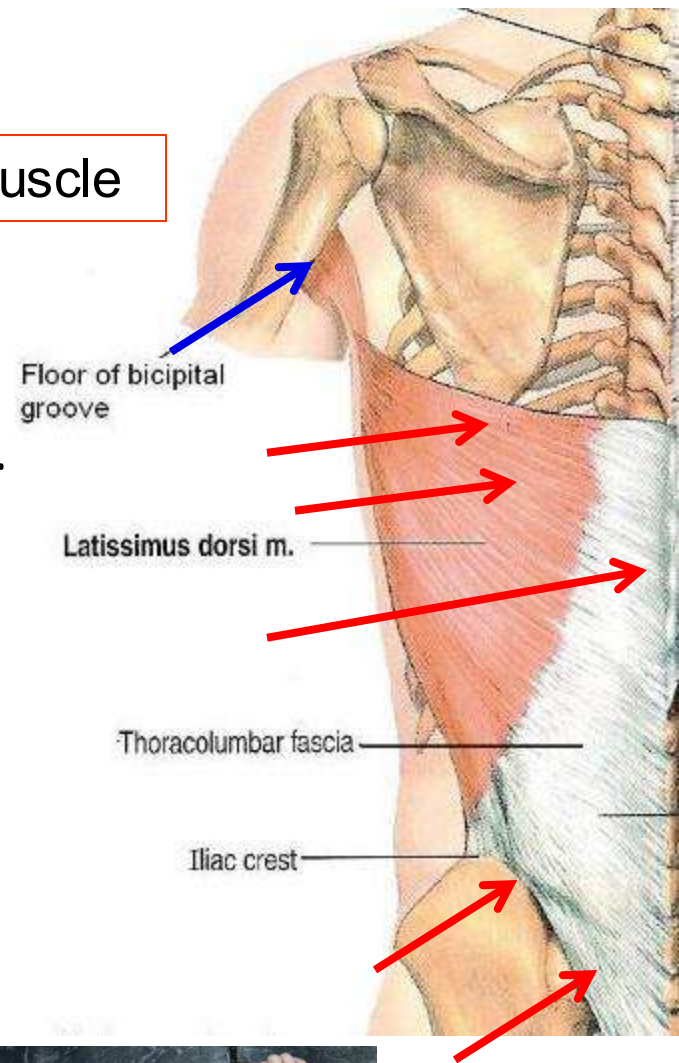
Thoracodorsal nerve.

Action:

Adduction, Extension

Medial rotation of the arm.

Helps in Climbing (**climbing muscle**)



Levator Scapulae

Origin:

The transverse processes of the UPPER four cervical vertebrae.

Insertion:

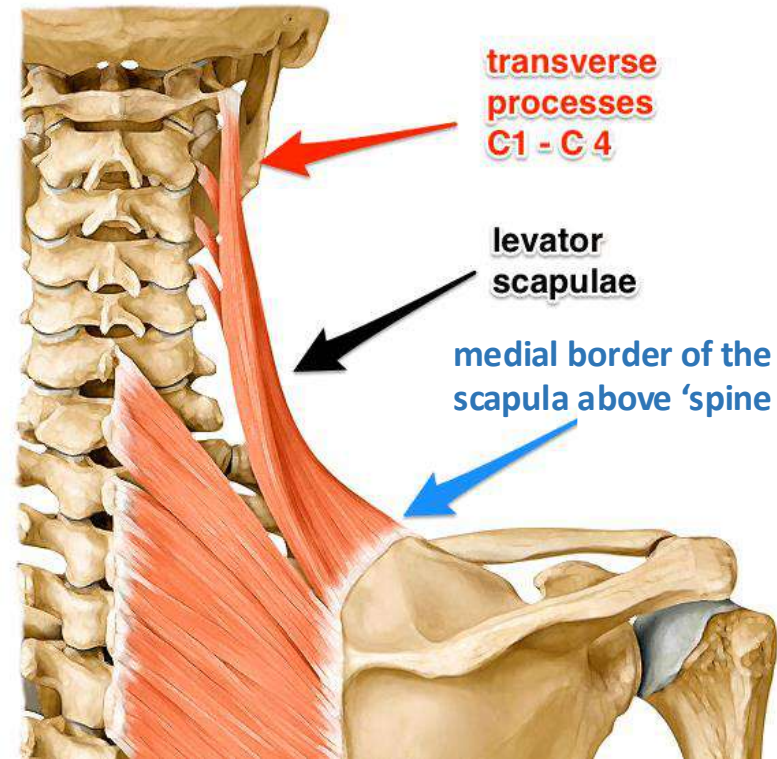
The medial border of the scapula **above** 'spine. (opposite to suprasp. fossa)

Nerve Supply:

From the C3 and C4 cervical ns. & from dorsal scapular nerve

Action:

- ✓ **Elevation** of the scapula (shoulder).
- ✓ **Help in Retraction** of the scapula (Pulls the scapula medially).
- ✓ **Lateral flexion of neck** It bends the head to the **ipsilateral** side.



Rhomboid Minor

Origin:

1. Lower part of ligamentum nuchae.
2. Spines of **C7 & T1**.

Insertion:

The medial border of the scapula opposite to the root of its spine.

Rhomboid Major

Origin:

From spines & supraspinous ligaments of **T2---T5**.

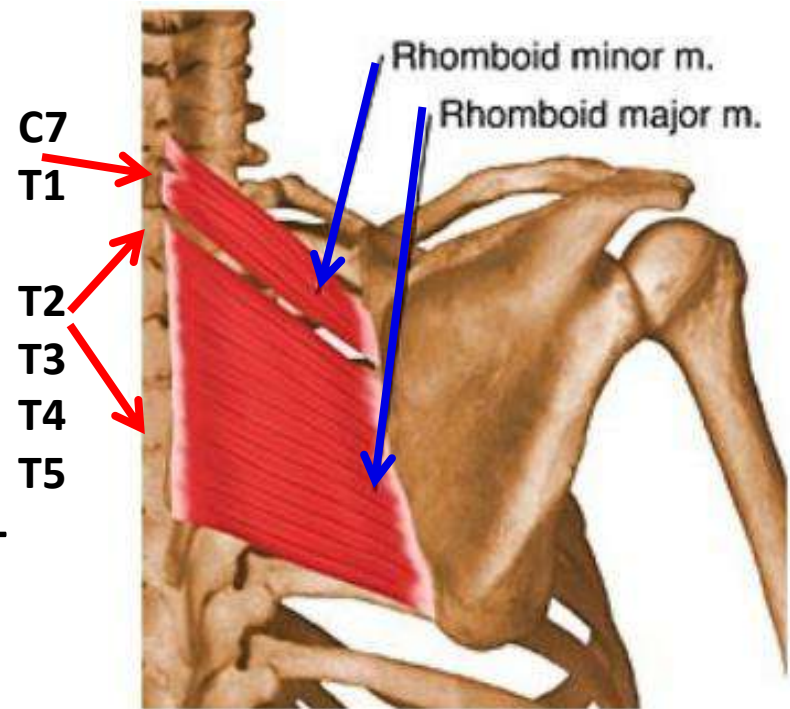
Insertion:

The medial border of the scapula **below** the spine. (opposite to infraspinatoues Fossa)

Nerve Supply: Dorsal scapular nerve.

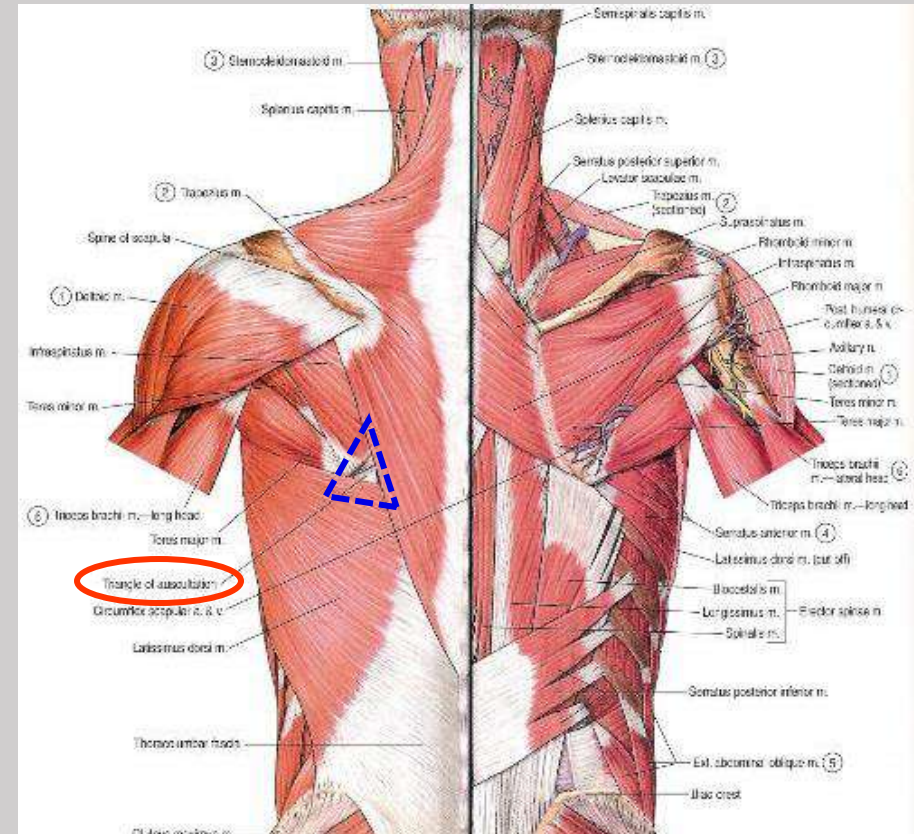
Action: - Elevation of the scapula.

- Retraction of the scapula. (It pull the scapula medially)



Triangle Of Auscultation

A good place to examine posterior segments of the lungs with a stethoscope



A triangular gap formed by:

- Laterally:** medial border of the **scapula**,
- Medially:** inferolateral border of the **trapezius**
- Inferiorly:** upper border of **latissimus dorsi**,

Muscles of the Shoulder regions

Deltoid

Origin:

Anterior fibers from the lateral 1/3 of anterior border of 'clavicle.'

Middle fibers from the lateral border of acromion process of 'scapula.'

Posterior fibers from the lower border of spine of 'scapula.'

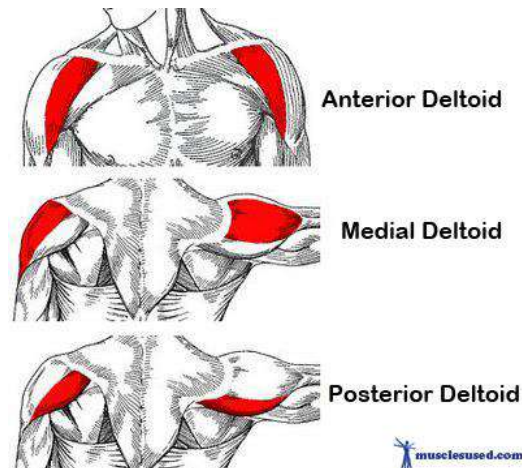
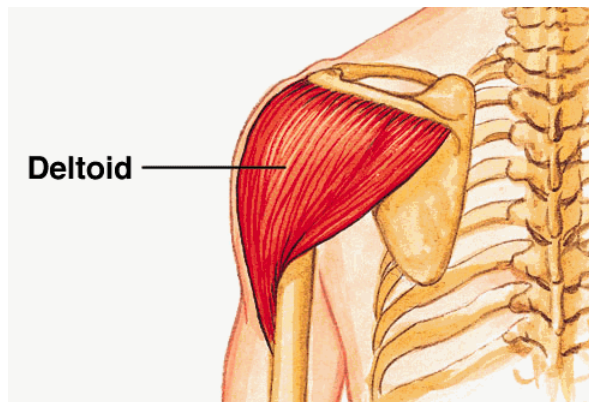
Insertion:

Middle of lateral surface of humerus (**deltoid tuberosity**)

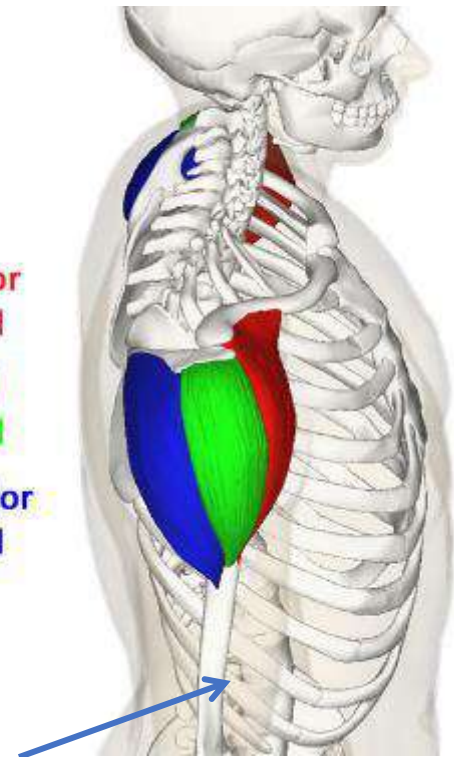
Nerve supply: **Axillary Nerve**

Action:

- Anterior fibers: **flexes & medially** rotate the arm
- Middle fibers: **Abducts** the arm (from **15 to 90°**)
- Poster. fibers: **extends & laterally** rotate the arm



Anterior
Deltoid
Lateral
Deltoid
Posterior
Deltoid



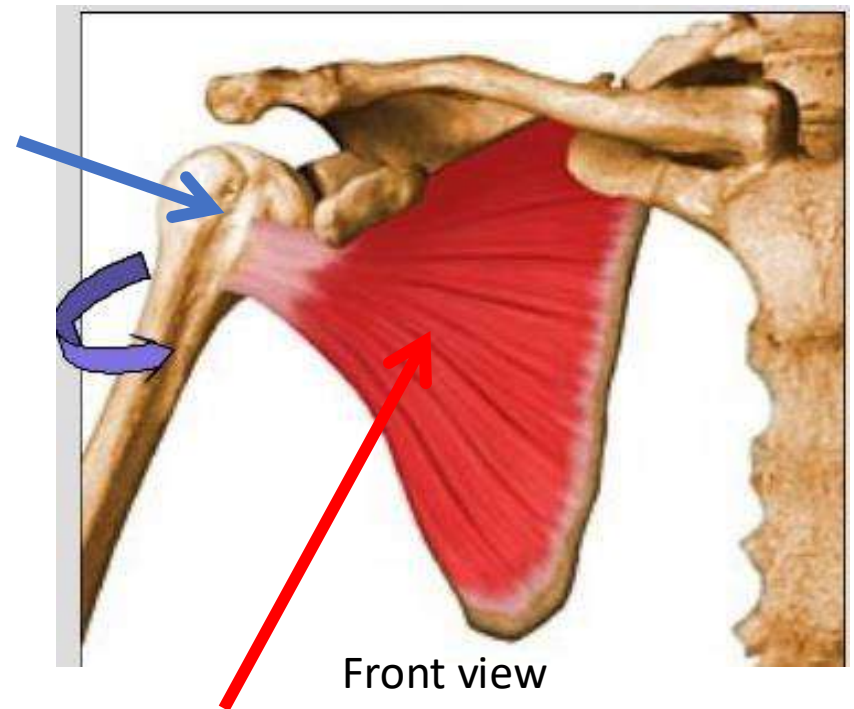
Muscles of the Scapular regions

Subscapularis

Origin: from the **Subscapular** fossa
Insertion: Lesser tuberosity of **humerus**
Nerve supply:
Upper & Lower Subscapular Nerves

Action:

- Medially rotates the arm
- Stabilizes the shoulder joint.



Supraspinatus

Origin: Supraspinous fossa of 'scapula

Insertion:

Greater tuberosity of humerus (tip);
Capsule of 'shoulder joint.

Nerve supply: Suprascapular Nerve

Action:

- ~ **Abducts** the arm (from 0° upto 15°)
- ~ Stabilizes the shoulder joint.

Infraspinatus

Origin: Infraspinous fossa of 'scapula

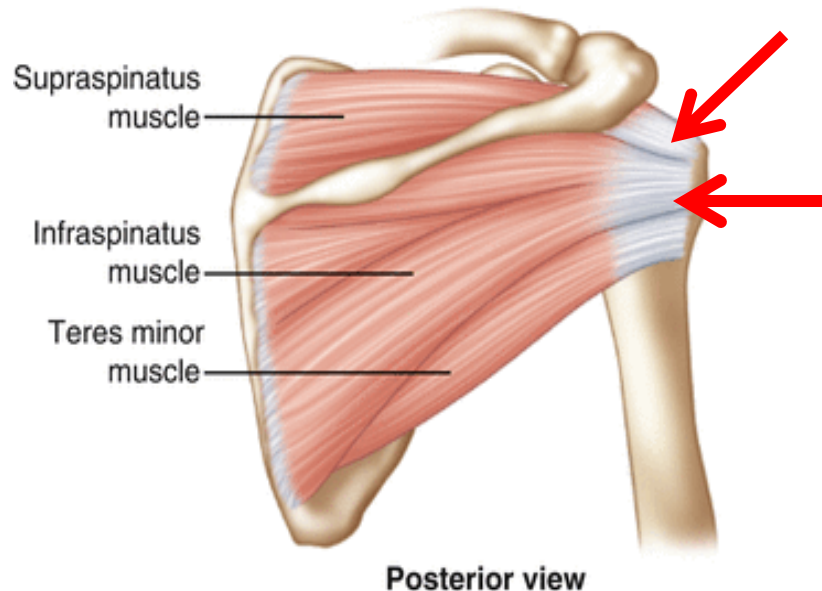
Insertion:

Greater tuberosity of humerus;
Capsule of 'shoulder joint.

Nerve supply: Suprascapular Nerve

Action:

- ~ **Laterally** rotates the arm
- ~ Stabilizes the shoulder joint.



Teres minor

Origin: Upper 2/3 of lateral border of 'scapula

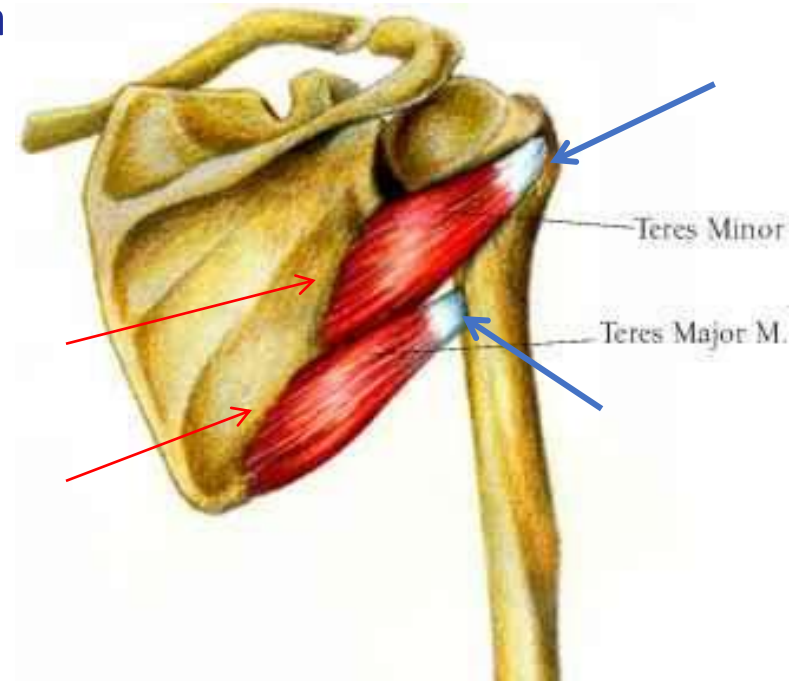
Insertion:

Greater tuberosity of humerus;
Capsule of 'shoulder joint.

Nerve supply: Axillary Nerve

Action:

- Laterally rotates the arm
- Stabilizes the shoulder joint.



Teres Major

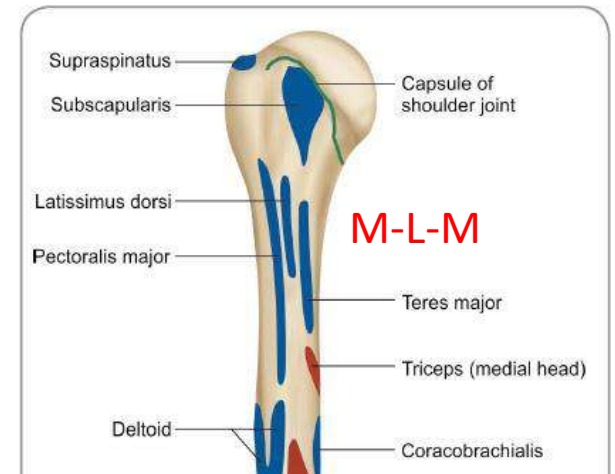
Origin: Lower 1/3 of lateral border of 'scapula

Insertion: Medial lip of bicipital groove of humerus.

Nerve supply: Lower Subscapular Nerve

Action:

- Adduction of the arm
- Medially rotates the arm



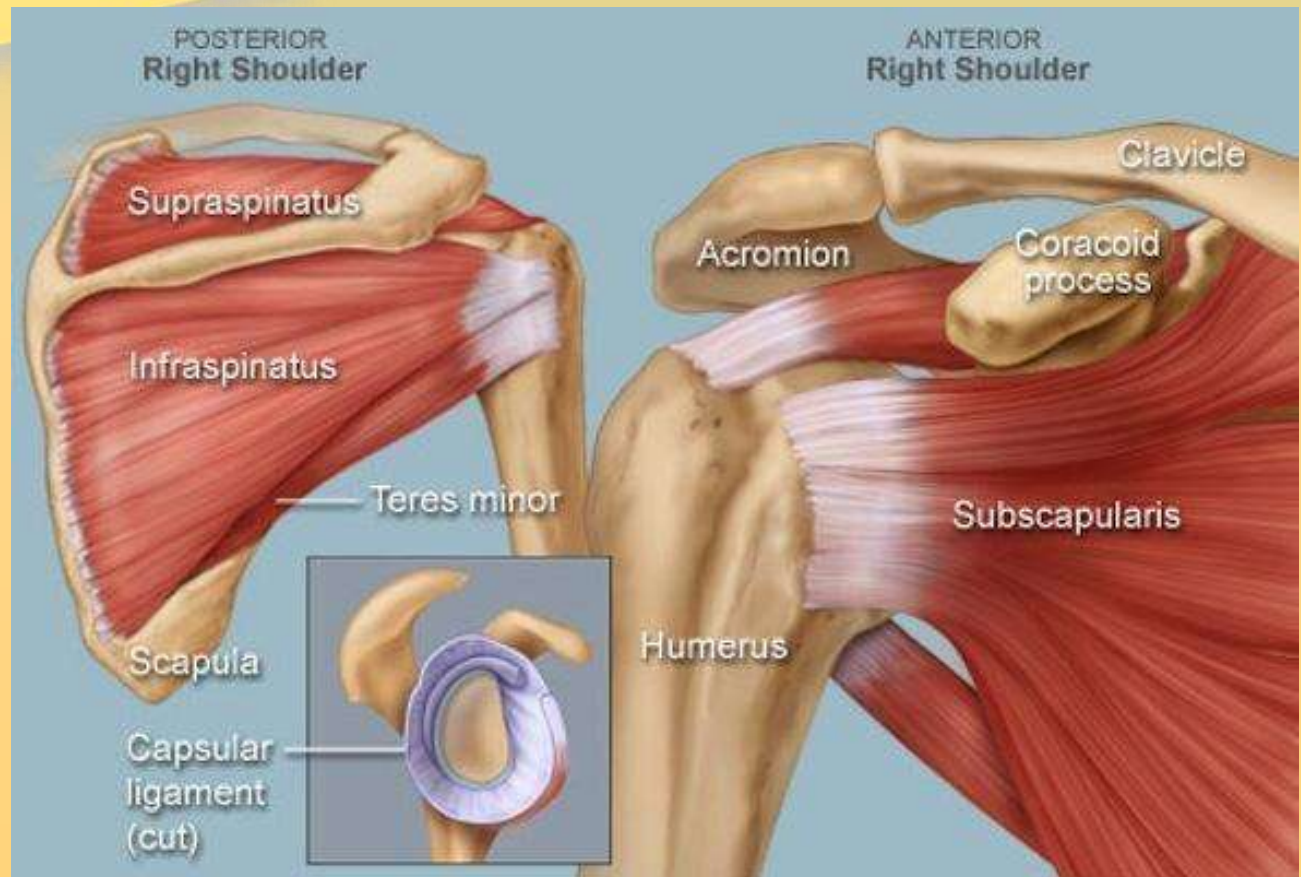
All ms inserted here--> medially rotate the arm

Muscles connecting 'scapula to 'humerus

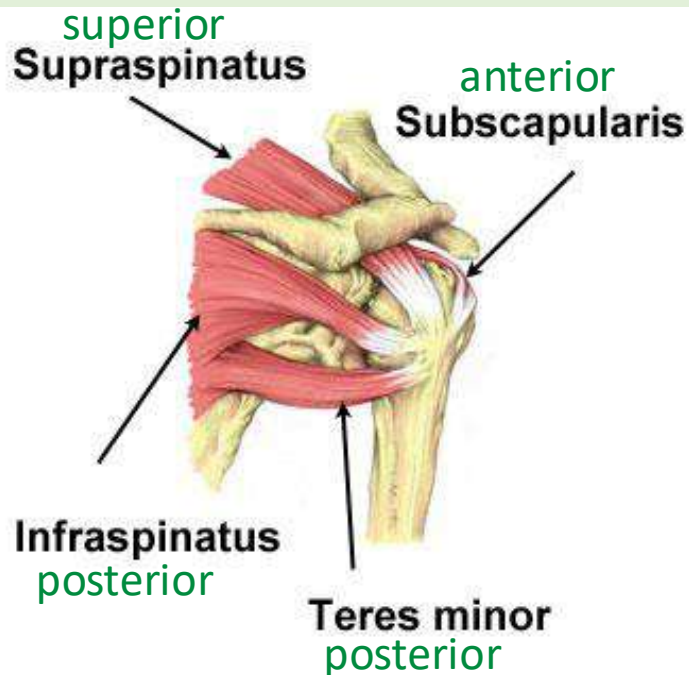
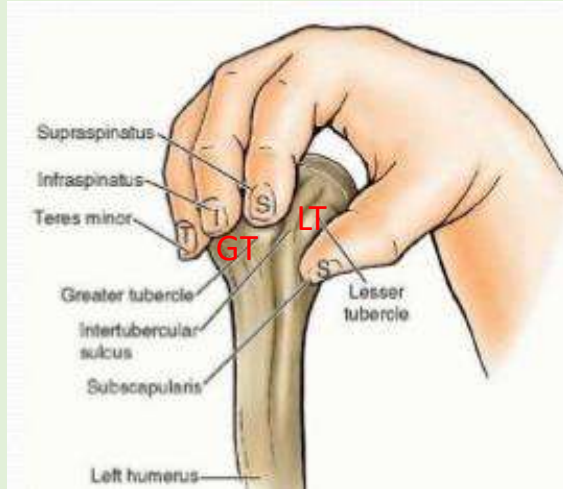
	ORIGIN	INSERTION	Nerve Supply	ACTION
Subscapularis	Subscapular fossa	Lesser tuberosity of H. Capsule of Sh.J	Upper & lower subscapular nerves	- Medially rotates arm - Stabilizes Sh.J
Supraspinatus	Supraspinous fossa	Greater (tip of) tuberosity of H. Capsule of Sh.J	Suprascapular nerve	- Abducts arm - Stabilizes Sh.J
Infraspinatus	Infraspinous fossa	Greater tuberosity of H. Capsule of Sh.J	Suprascapular nerve	- Laterally rotates arm - Stabilizes Sh.J
Teres minor	Upper 2/3 of lateral border of scapula	Greater tuberosity of H. Capsule of Sh.J	Axillary nerve	- Laterally rotates arm - Stabilizes Sh.J
Teres major	Lower 1/3 of lateral border of 'scapula	Medial lip of bicipital groove of humerus	Lower subscapular nerve	- Medially rotates & adducts arm
Deltoid	Clavicle, Acromion, Spine	Deltoid tuberosity of humerus	Axillary nerve	Many actions !!

Remember!

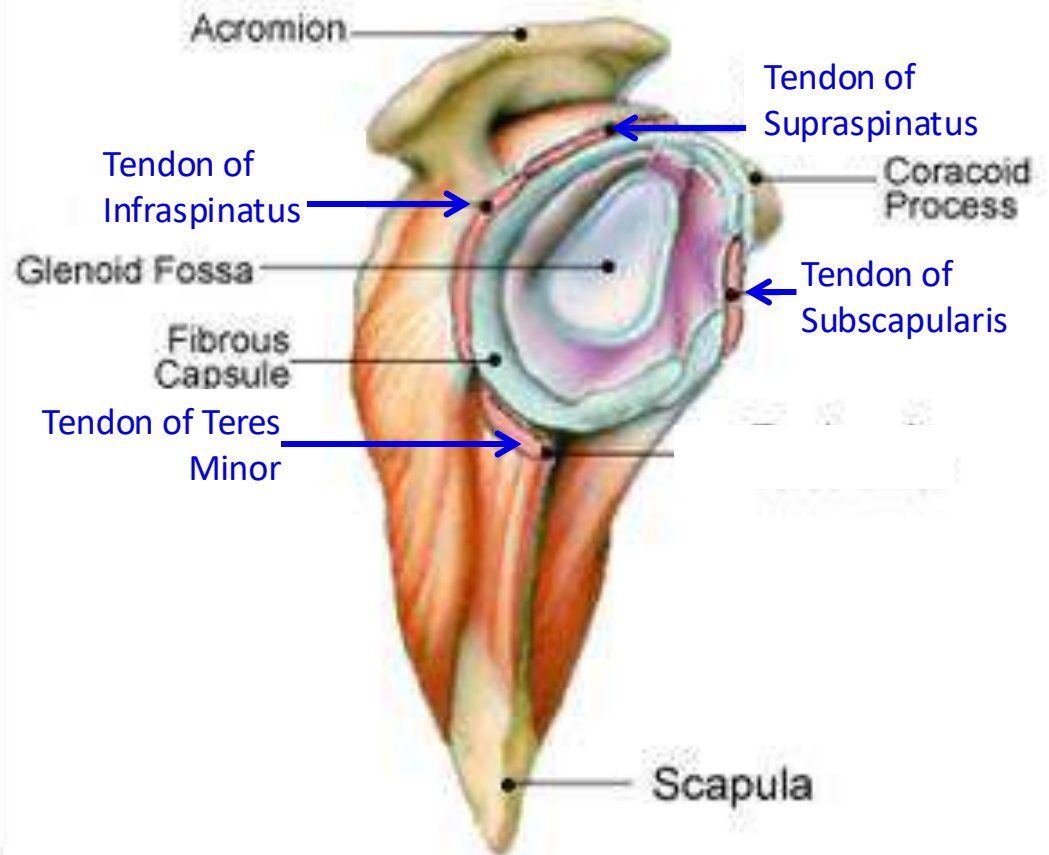
Scapular muscles



The Rotator Cuff



Rotator Cuff Muscles



The Rotator Cuff

The rotator cuff is the name given to the **tendons of muscles** (which are fused to the capsule of the shoulder joint to stabilize it during its movement).

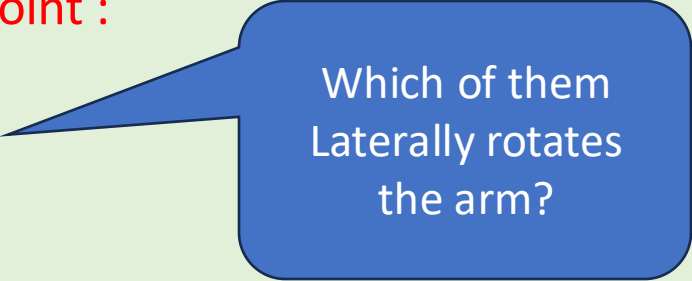
Their tone helps in holding the head of humerus in the glenoid cavity.

Muscles:

Subscapularis
Supraspinatus
Infraspinatus
Teres minor

They surrounds the joint :

anteriorly
superiorly
posteriorly
posteriorly



Which of them
Laterally rotates
the arm?

So, the cuff is deficient inferiorly → this is 'site of potential weakness

Note: The common shoulder dislocation occurs...

Inferiorly WHY?

Which nerve can be injured ?

The axillary nerve WHY?

Arterial Anastomosis Around The Scapula & Shoulder Joint

<https://www.youtube.com/watch?v=D3NuZX8XgMQ&t=166s>

Clinical importance:

It ensures adequate blood flow to the upper limb during the movement of the arm, to compensate the compression of 'axillary artery.

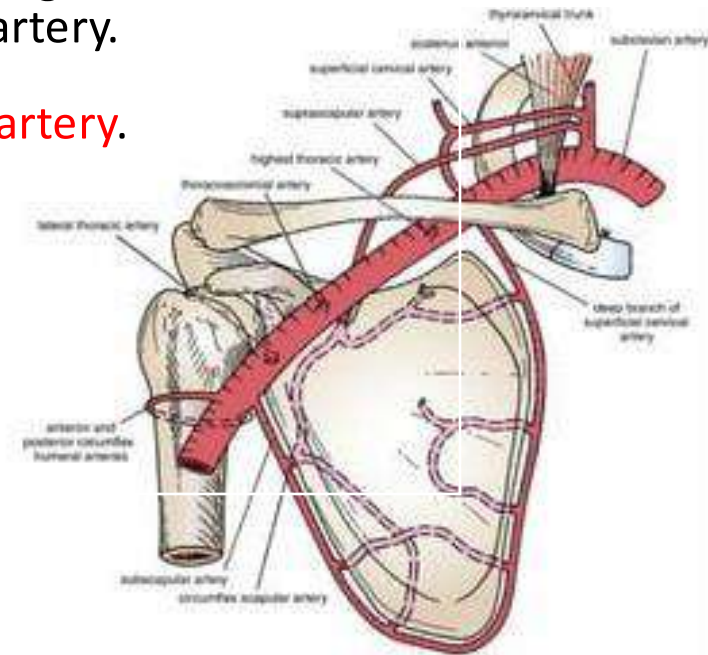
So, there is an arterial **anastomosis between:**

the **subclavian artery** & the **axillary artery**.

Where??

Around the scapula & acromion

Around the surgical neck of 'humerus



The **subclavian artery** and the **axillary artery** give branches to form the scapular and acromial anastomoses .

The different **segments** of the **axillary artery** give branches to form the shoulder joint anastomoses.

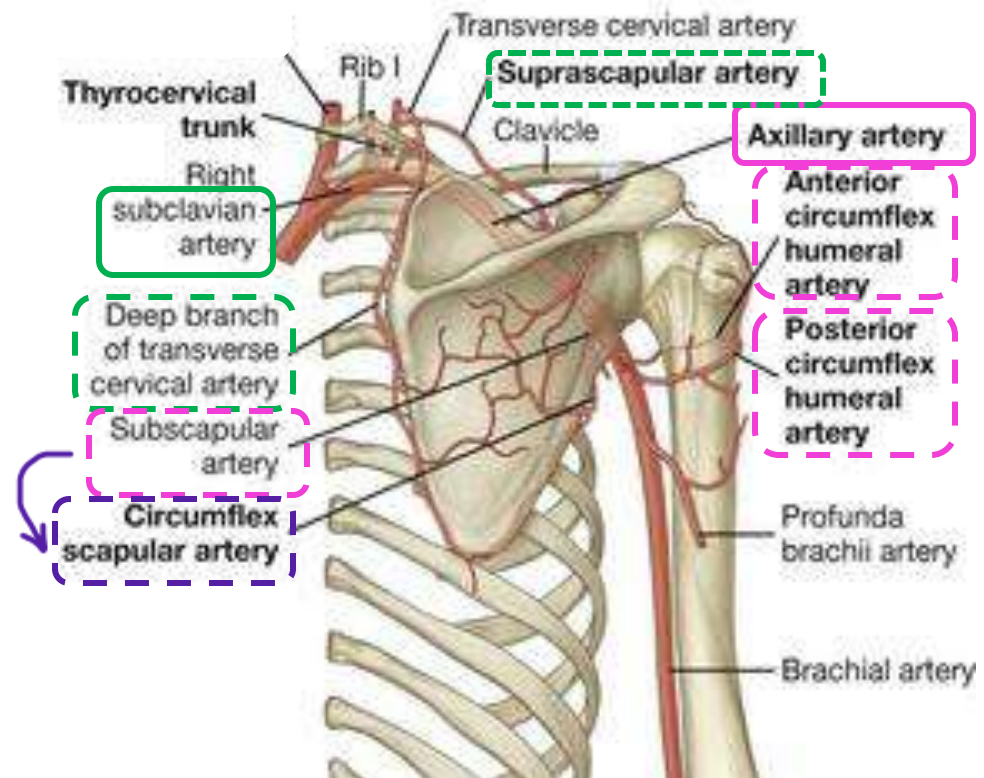
Arterial Anastomosis Around The Scapula & Shoulder Joint

branches from the subclavian artery:

- a) Suprascapular artery runs on supraspinous & infraspinous fossae.
- b) Deep branch of transverse cervical artery runs on the medial border of 'scapula.

branches from the axillary artery: (part ???)

- a) Subscapular artery runs on lateral border of the scapula & gives off
circumflex scapular artery runs on infraspinous fossa
- b) Posterior circumflex humeral artery,
- c) Anterior circumflex humeral artery



The Nerves ..

Suprascapular nerve

Axillary nerve

Suprascapular nerve

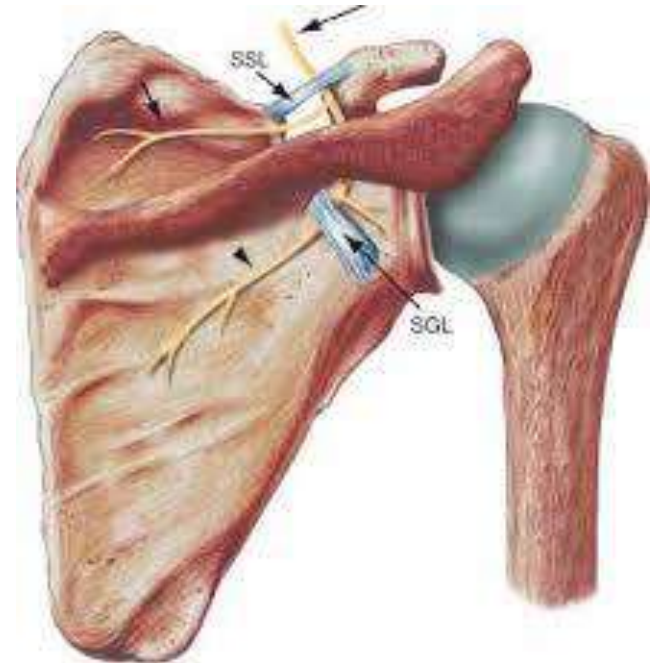
Begins:

arises from the **upper trunk** of the brachial plexus.

It runs downward and laterally & passes through suprascapular notch to reach 'supraspinous fossa.

Supplies what ?

- ~Articular → the shoulder **joint**,
- ~Muscular → two **muscles**
(supraspinatus & infraspinatus).



Axillary nerve

Begins:

In the axilla, arises from the **posterior cord** of 'brachial plexus.

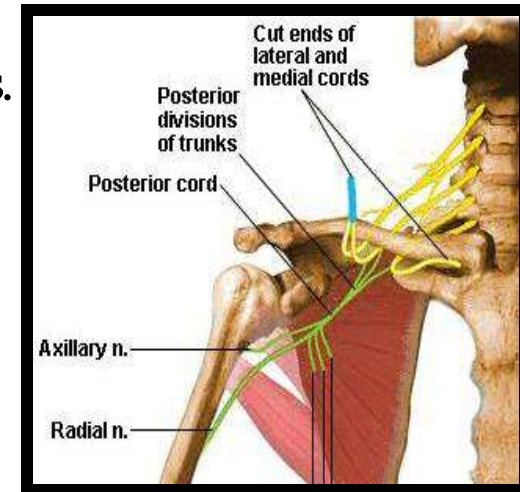
Course:

It passes backward **with** 'posterior circumflex humeral artery

The nerve is **related** to the **surgical neck** of the humerus and passes below the **shoulder** joint.

Ends:

by dividing into anterior & posterior terminal branches.



Branches

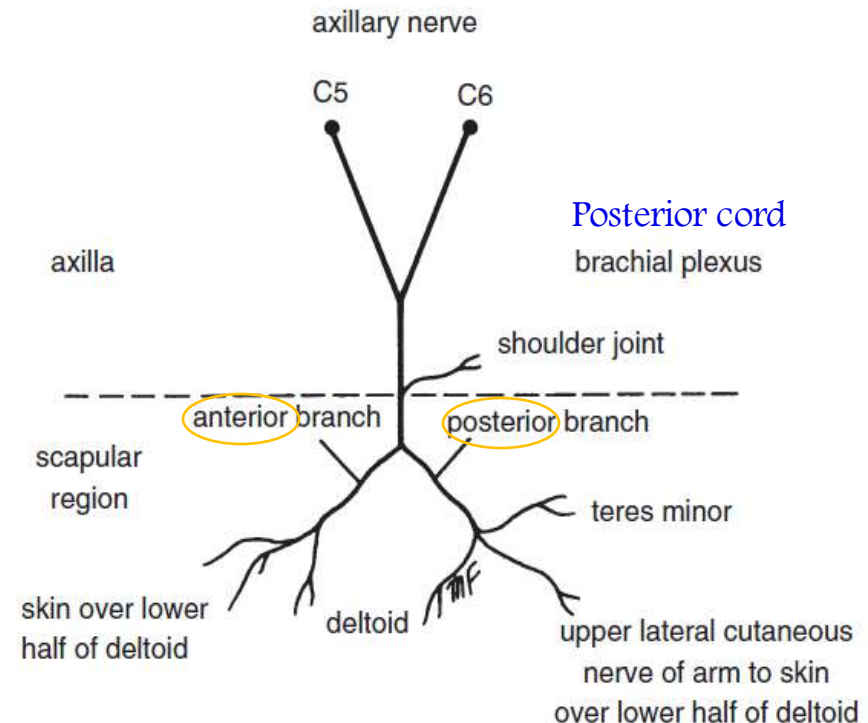
■ **Articular branch:** to the **shoulder joint**

■ **Anterior terminal branch:**

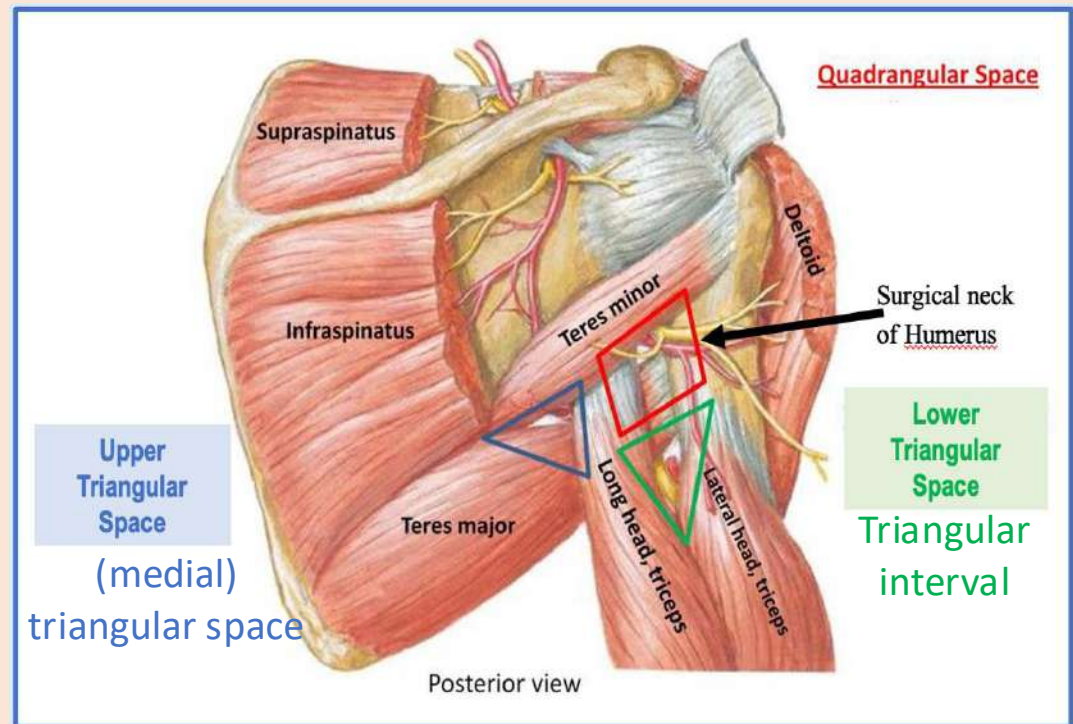
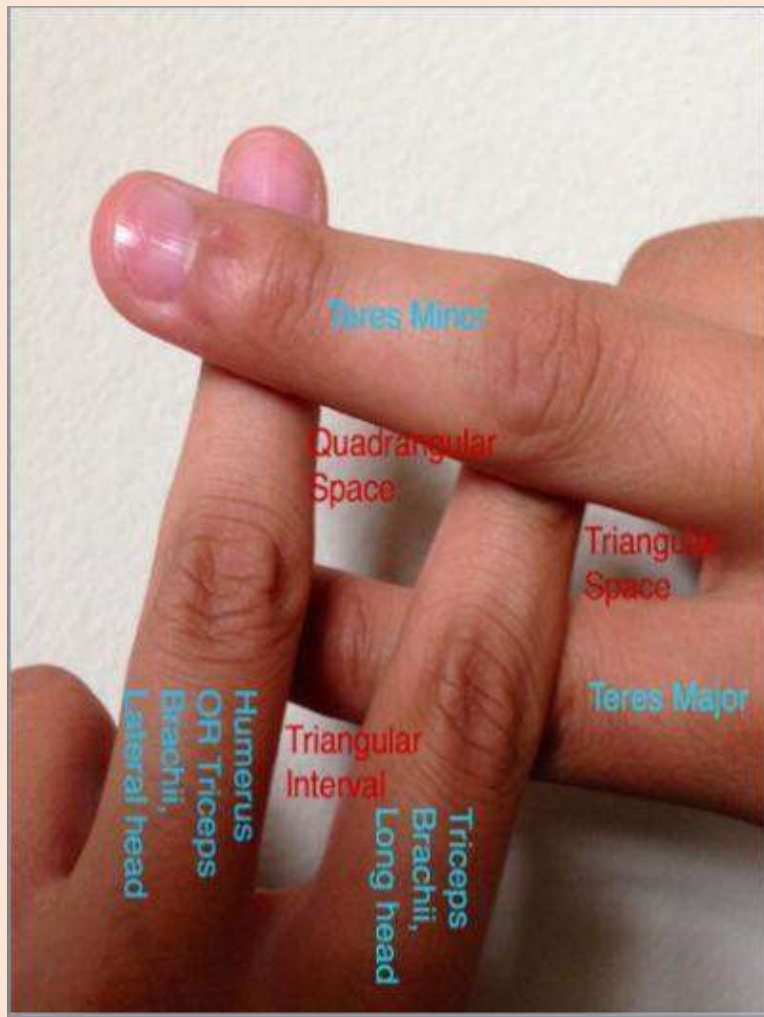
It supplies: the **deltoid**,
the **skin that covers its lower part of deltoid**.

■ **Posterior terminal branch:**

It supplies: the **deltoid** and **teres minor** ,
It ends as the **upper lateral cutaneous nerve of the arm**.



Intermuscular Spaces



Quadrangular Space

Boundaries:

Superiorly:

subscapularis (anter.)
teres minor (poster.)

Inferiorly:

teres major

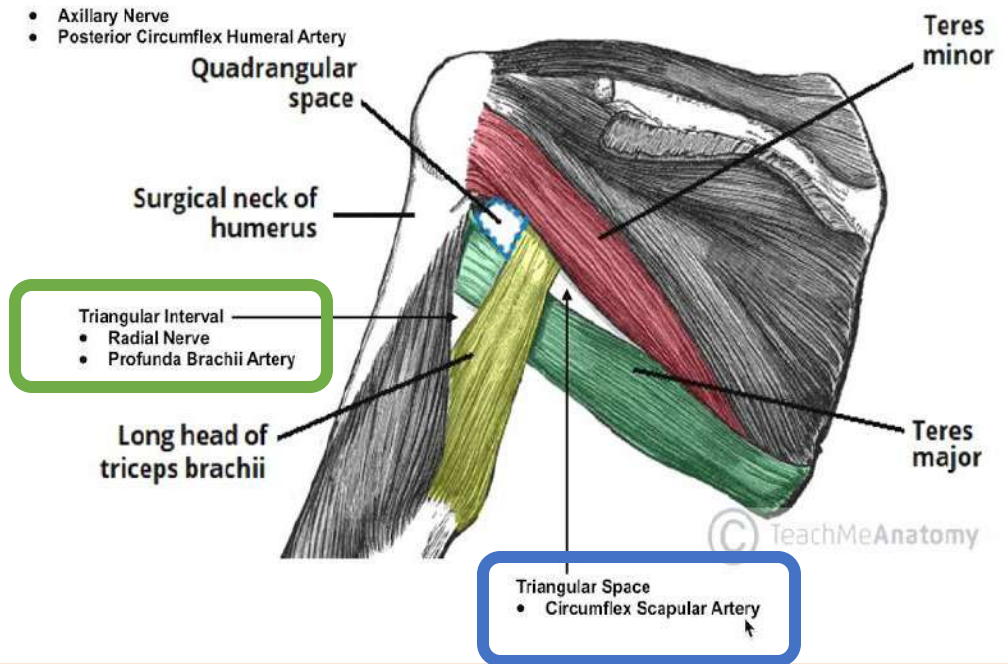
Medially:

long head of triceps.

Laterally:

lateral head of triceps &
surgical neck of humerus

The Quadrangular and Triangular Spaces; The Triangular Interval

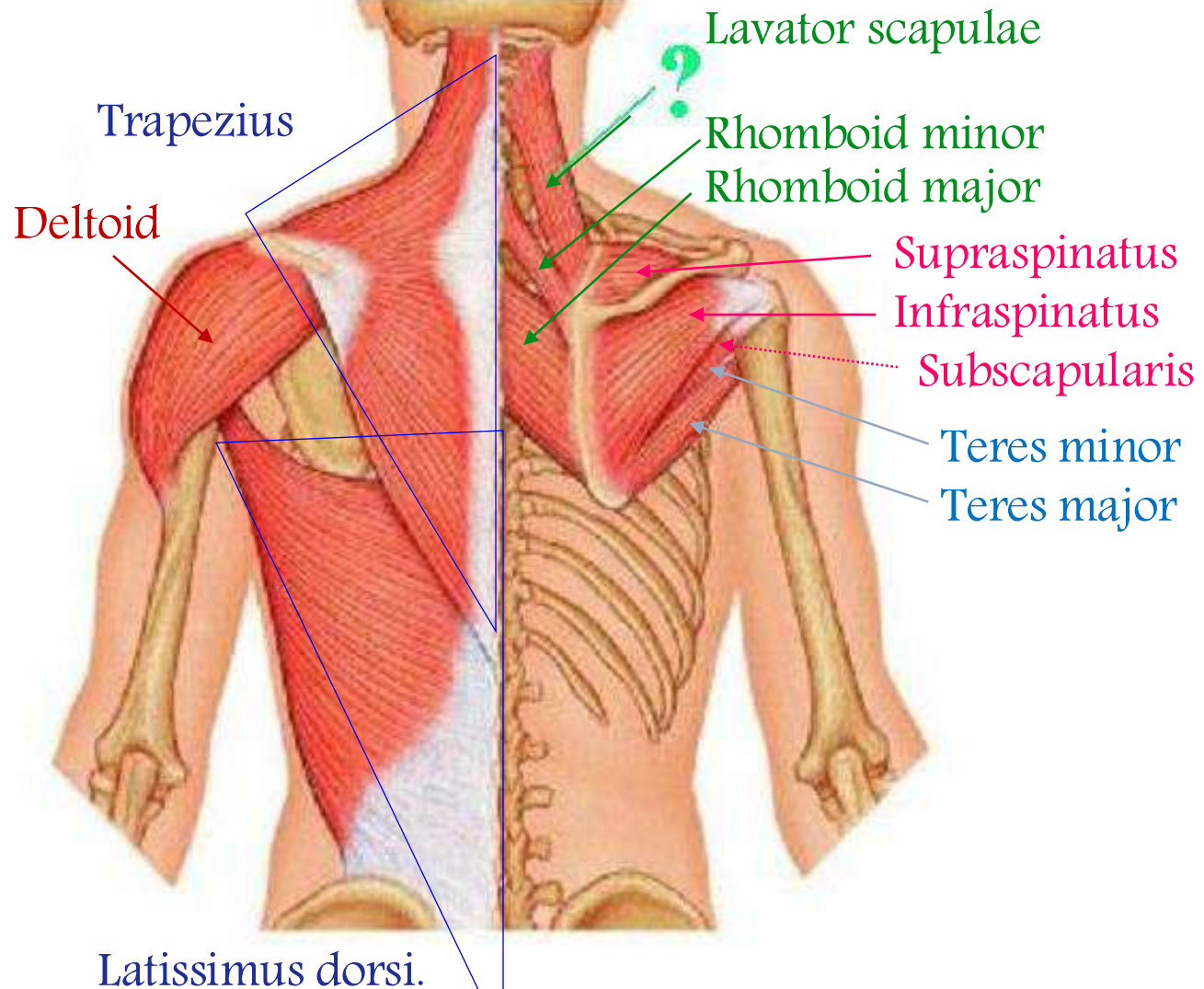


Contents:

Axillary nerve.

Posterior circumflex humeral vessels.

Scapular, Shoulder & Back Regions





Clinical Notes

A man 45 years old complaining of shoulder pain and he cannot raise his arm by himself. His job is fixing the lightening in high ceilings and cleaning windows everyday.



What happened to him?

Tearing of the Rotator Cuff

Tearing of the rotator cuff tendons is a **painful** injury.
A torn rotator cuff creates a very **weak** shoulder and affects its **movement**.

Causes:

The areas of poor blood supply in the rotator cuff make these tendons at risk to degeneration from aging.

It is a common injury later in life or with overusing shoulder joint.

Who can be affected ?

Sports such as baseball & Basketball players, swimmers & Kayak sports.

Routine daily works like cleaning windows, washing and waxing cars,

Painting can lead to rotator cuff fatigability due to overuse.

Excessive force like trying to catch a heavy falling object OR a fall directly on shoulder.





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